Geochemistry Geophysics Geosystems G3

Article
Volume 12, Number 6
24 June 2011
Q06010, doi:10.1029/2011GC003570
ISSN: 1525-2027

Published by AGU and the Geochemical Society

Curvature of subducted lithosphere from earthquake locations in the Wadati-Benioff zone

B. A. Buffett
Department of Earth and Planetary Science, University of California, Berkeley, California 94720, USA (bbuffett@berkeley.edu)

A. Heuret
Dipartimento Scienze Geologiche, Università degli Studi "Roma Tre," Largo S. L. Murialdo 1, I-00146 Rome, Italy

[1] Estimates of slab curvature from 133 profiles across 12 major subduction zones are obtained by fitting splines through earthquakes locations. Several geometric quantities are computed from the spline fits, including the minimum radius of curvature. We also introduce a quantity called the equivalent radius of curvature to precisely define the bending dissipation in either a viscous or a plastic slab. The minimum and equivalent radius of curvature are approximately equal in slabs with simple geometry (i.e., a single bend) but deviate substantially when the geometry is more complicated. A compilation of estimates shows that the equivalent radius of curvature is insensitive to the age of the plate at the trench. Mean values are 175 km or 191 km, depending on the rheology of the slab. When bending occurs by viscous deformation, the resulting dissipation can exceed the power available from subduction with reasonable choices for viscosity. A more consistent description of the energetics is obtained by assuming a plastic rheology.

Components: 6100 words, 6 figures, 1 table.

Keywords: dissipation; rheology; subduction.

Index Terms: 7230 Seismology: Seismicity and tectonics (1207, 1217, 1240, 1242); 8159 Tectonophysics: Rheology: crust and lithosphere (8031); 8170 Tectonophysics: Subduction zone processes (1031, 3060, 3613, 8413).

Received 18 February 2011; **Revised** 20 April 2011; **Accepted** 27 April 2011; **Published** 24 June 2011.

Buffett, B. A., and A. Heuret (2011), Curvature of subducted lithosphere from earthquake locations in the Wadati-Benioff zone, *Geochem. Geophys. Geosyst.*, *12*, Q06010, doi:10.1029/2011GC003570.

1. Introduction

[2] Earthquake locations in Wadati-Benioff zones are often used to delineate subducted slabs [e.g., *Stauder*, 1973; *Barazangi and Isacks*, 1976]. Regions with sufficient seismicity permit detailed reconstructions of slabs in three dimensions [*Chiu et al.*, 1991; *Cahill and Isacks*, 1992; *Pardo and Suarez*, 1995]. Quantitative estimates of curvature [*Bevis and Isacks*, 1984; *Bevis*, 1986; *Heuret*, 2005] and deformation [*Boyd and Creager*, 1991; *Nothard et al.*, 1996] can be computed by fitting surfaces through the earthquake locations. The strains required to accommodate changes in shape during subduction indicate that slabs undergo both bending and in-plane deformation.

[3] In this study we use hypocenters from *Engdahl et al.* [1998] to construct two-dimensional profiles of slabs at locations defined by *Heuret and Lallemand* [2005]. The goal is to compute the radius of curvature and related quantities that define the bending dissipation in slabs [*Becker et al.*, 1999;

Copyright 2011 by the American Geophysical Union

Conrad and Hager, 1999]. All of these quantities are derived from estimates of the local slab dip, $\theta(s)$, as a function of distance, s, along the slab. The curvature of the slab is defined by

$$K(s) = \frac{d\theta}{ds} \tag{1}$$

and the radius of curvature is given by $R(s) = K(s)^{-1}$. Deformation due to bending depends on changes in $K(s)$ along the slab (i.e., dK/ds).

[4] Estimates for $\theta(s)$ and derivatives are obtained by fitting a cubic smoothing spline through earthquake locations in the slab. Our calculations reveal that the minimum radius of curvature is nearly independent of the age (or thickness) of the subducting plate at the trench. By comparison laboratory [*Bellahsen et al.*, 2005] and numerical [*Capitanio et al.*, 2009] studies show that the radius of curvature in viscous slabs increases with plate thickness when other properties are held constant (e.g., density, width, viscosity). Maintaining a constant radius of curvature in slabs with increasing thickness implies large variations in strain rate and imposes constraints on the rheology required to achieve those strain rates. We explore this question from an energetic point of view in section 4. For the moment it suffices to note that the gravitational energy available from cold, dense slabs can be insufficient to overcome the dissipation due to bending for plausible values of viscosity. We find that a simple plastic model reduces these difficulties by altering the dependence of dissipation on the subduction velocity. When bending occurs by plastic deformation, the resulting dissipation accounts for no more than 40% of the buoyancy flux due to slabs in the upper mantle.

2. Spline Fit to Hypocenters

[5] We use cubic splines to fit hypocenters on a series of profiles that are perpendicular to the subduction boundary. A typical profile includes events within roughly one hundred kilometers of a central transect. The location and width of each profile are listed in Table 1. Earthquake locations are specified by depth, z, and horizontal position, x, along the transect. A spline representation of the hypocenters (denoted by $z_s = f(x)$) is obtained by minimizing the misfit $|z - z_s|^2$, subject to a penalty on the roughness of the fit, measured by the second derivative $(d^2f/dx^2)^2$. A smoothing parameter, p, defines the relative weighting of the misfit and the penalty in the minimization. Setting $p = 0$ gives the least squares fit of a straight line through the hypocenters, whereas $p = 1$ gives a conventional spline fit. (The smoothing spline is implemented using MATLAB function csaps).

[6] Figure 1 shows several fits to the hypocenters from profile C_ALE1 in the central Aleutians [see *Wu et al.*, 2008]. Different choices for the smoothing parameter yield somewhat different results. The lowest value for p allows the smallest curvature and produces the largest the misfit to the hypocenters, particularly at shallow depths. Better fits are obtained as p is progressively increased. The largest value for p in Figure 1 gives the best fit, although all three fits have insufficient curvature to pass through the center of seismicity at depth. This result suggests that even the highest value for p is still too small. The estimate for the minimum radius of curvature (denoted by $R_{\min}$) is liable to be too large. Qualitatively similar results are obtained at the other subduction zones, so we make a conservative estimate for the curvature and bending dissipation by using $p = 10^{-6}$ for all subduction zones.

[7] Most previous studies define the top of the slab by fitting a surface through the upper envelope of hypocenters. Shallow earthquakes ($z < 50$ km) occur primarily at or near the interface with the overlying plate, whereas intermediate and deep focus events are thought to occur in the cold interior of the slab [*Molnar et al.*, 1979]. We fit splines through the center of the seismicity with the aim of approximating the mechanical centerline of the slab, commonly used to describe the dynamics of a thin plate. This approach can yield a centerline which is too shallow at the trench and outer rise because most of the seismicity in this region is located near the top of the slab (or plate). A best fitting spline is expected to deviate from the centerline near the outer rise, causing a small decrease in curvature at the bend (i.e., an increase in the radius of curvature). On the other hand, the depth of the preferred spline fit through the outer rise ($x < 0$ km in Figure 1) is only slightly shallower than the 25 km depth cutoff for earthquakes under tension on the outer rise [*Forsyth*, 1982]. A transition to compressional events (necessarily below the centerline) brackets the appropriate location of the spline fit. Translating the spline to a depth of (say) 30 km on the outer rise might give a better approximation to the centerline. It would also improve the fit to the seismicity at depth, but it would not change the curvature.

[8] A larger source of uncertainty is due to the choice of smoothing parameter. The range of values considered in Figure 1 correspond to a 40% variation in the estimate of $R_{\min}$, although the lowest value for

Table 1. Subduction Profiles Included in This Study[a]

Zone	Profile	Latitude	Longitude	Width (km)	Azimuth (deg)	R_{min} (km)	R_{vs} (km)	R_{pl} (km)
AlAl	CALE1	50.8	177	137.7	20.9	143	131	143
	CALE2	50.4	179	129.6	11.6	131	124	131
	CALE3	50.3	181	144.3	350.9	170	167	164
	CALE4	50.4	183	128.8	350.2	98	89	99
	CALE5	50.5	185	129.4	352	133	133	134
	CALE6	50.9	187	127.6	353	168	159	165
	EALE1	51.1	189	130.6	342.3	190	178	192
	EALE2	51.5	191	133	336.2	166	160	164
	EALE3	52.0	193	132.3	337.9	134	136	129
	EALE4	52.5	195	132.5	338.0	200	166	203
	EALE5	53.1	197	128.4	339.5	145	125	142
	EALA1	56.2	209	123.7	331.0	468	206	262
	EALA2	57.1	211	146.1	319.6	391	261	333
	EALA3	58.0	213.0	150.6	314.5	258	203	261
	EALA4	59.1	214.5	159	310.6	341	148	136
	EALA5	59.2	215.5	117.9	345	419	210	346
	WALA1	53.5	199	124.5	343.5	221	189	210
	WALA2	53.8	201	124	341.6	243	229	246
	WALA3	54.2	203	123.5	341.3	330	252	327
	WALA4	54.8	205	138.9	323.1	292	220	252
	WALA5	55.6	207	127.1	330.1	369	312	369
	WALE2	51.1	175	130.3	16.2	159	144	163
Ant	ANTI1	16	301.2	230.2	250	250	218	247
	ANTI2	18	299.8	314.5	230	263	227	265
	ANTI3	19.3	298	223.4	205	321	267	317
	PORTO1	19.8	296	210	185	212	173	213
	PORTO2	19.8	294	210	176	251	209	258
CA	COST1	14.2	266	225.4	21.0	307	177	177
	COST2	13.2	268	219.5	26.5	229	157	158
	COST3	12.3	270	211.1	22.2	164	167	165
	COST4	11.4	272	233.8	32	106	108	106
	COST5	9.4	274	219.4	30	132	133	132
	MEX1	19.1	254.5	328.2	54.9	276	212	280
	MEX2	17.9	256	232.7	37.4	300	168	208
	MEX3	16.9	258	212.8	25	402	178	247
	MEX4	16.3	260	204.9	30.4	343	174	218
	MEX5	15.6	262	202.1	17.8	579	226	297
	MEX6	15.3	264	196.5	9.7	290	207	239
JK	IZU1	33	142.1	202.3	260.9	200	145	116
	IZU2	31	142.3	205.3	258	209	146	100
	IZU3	29	142.9	210.4	252.3	167	120	90
	IZU4	27	143.3	222.4	272	116	114	83
	JAP1	40.5	144.5	216.5	291.9	231	119	95
	JAP2	39	144.2	200.3	276.6	286	196	250
	JAP3	37	143.5	226.8	298.9	319	258	319
	JAP4	35	142.2	217.9	297.1	188	115	103
	KAM1	54.5	163.6	256.8	300	149	137	136
	KAM2	53.0	162.3	222.4	300	219	196	205
	NKOUR1	51.0	160.2	237.9	304.1	234	204	222
	NKOUR2	49	157.5	282.5	313.9	212	182	208
	NKOUR3	47	155.1	251.2	307.6	144	128	115
	SKOUR1	45.3	153	195.2	318.3	178	156	166
	SKOUR2	44.4	151	163.9	329.4	161	144	136
	SKOUR3	43.5	149	191.1	318.9	155	120	130
	SKOUR4	42.2	147	191.2	324	169	141	148
	SKOUR5	41.5	145.5	207.3	318.5	254	159	190

Table 1. (continued)

Zone	Profile	Latitude	Longitude	Width (km)	Azimuth (deg)	R_{min} (km)	R_{vs} (km)	R_{pl} (km)
M	NMAR1	23	145.1	253.3	253.3	208	185	189
	NMAR2	21	146.9	314.5	235	179	166	154
	NMAR3	19	147.6	236.7	260	154	131	123
	NMAR4	17	147.8	222.4	271	102	107	97
	SMAR1	15	147.5	211.8	289.2	130	106	89
	SMAR2	13.3	146.5	291	314.4	237	156	142
	SMAR3	12	144.5	212.9	335.9	106	107	104
NB	BOUG1	−6.5	154	255.2	35	74	81	63
	BOUG2	−8	156	254.4	33	129	127	129
	BRET1	−7.3	148	202.6	10	73	73	68
	BRET2	−7	150	243.6	339	103	108	103
	BRET3	−6.1	152	235.3	335	97	92	87
NH	ENTR	−16	166.6	256.8	64	88	92	80
	NHEB1	−14	166.2	230.2	83	102	93	67
	NHEB2	−12	165.7	283.8	66	87	86	87
	SHEB1	−22	169.5	291.5	44.6	141	121	142
	SHEB2	−20	168.2	224.2	61.4	109	112	104
	SHEB3	−18	167.4	200.4	81.5	113	114	109
RN	RYUN1	24.2	127	247.6	325	332	271	319
	RYUN2	25.7	129	231.4	320	226	167	170
	RYUN3	27.5	130.5	257.6	310	172	163	169
	RYUN4	29.8	132	251.9	300	168	161	168
	RYUS	23.4	124	204.2	345	172	170	173
SA	COL1	5	281.9	113	256.8	241	199	241
	COL3	3.5	280.9	290.3	120	261	196	271
	COL4	0	279	245.4	107	349	170	149
	COL5	−2	278.5	230.2	102	563	176	167
	JUAN1	−29	287.7	204.5	99.4	843	187	196
	JUAN2	−30.5	287.5	198.4	94.3	331	174	159
	JUAN3	−32	287.4	202.2	94.8	298	134	103
	NCHI1	−17	285.9	330.9	37.8	405	159	147
	NCHI2	−19	288	240	56.4	337	246	278
	NCHI3	−21	288.7	201	84.4	320	152	142
	NCHI4	−23	288.7	201.2	94.7	390	140	113
	NCHI5	−25	288.6	201.5	93.3	295	137	118
	NCHI6	−27	288.3	201.2	101.1	261	148	129
	PER1	−4	278.1	200.5	91	374	214	236
	PER2	−5.5	278.1	200	82.9	514	263	350
	PER4	−9	279.2	224.2	64	414	210	264
	PER6	−13	281.7	201.2	55.5	340	194	172
	PER7	−15	283.4	222.4	50	404	198	199
	SCHI1	−34	286.9	220	113.3	399	269	364
	SCHI2	−35.5	286.2	219.8	118.4	340	259	321
	SCHI3	−37	285.4	208.7	105	384	303	389
	SCHI4	−39	284.9	230.2	99	350	295	354
	SCHI5	−41	284.7	222.4	97	356	289	363
Sand	SAND2	−58	336.2	223.2	265	86	87	85
	SAND3	−56	334.9	290.3	240	75	79	69
	SAND4	−55.1	333	140.2	205	91	96	88
SU	JAVA1	−11.3	117	196.6	358.7	229	201	201
	JAVA2	−11.2	115	199.7	11.4	261	236	242
	JAVA3	−10.7	113	219.3	8	374	266	324
	JAVA4	−10.4	111	222.1	10	235	242	236
	JAVA5	−10.5	109	198.3	10	386	281	307
	JAVA6	−9.7	107	219	24.7	331	292	315
	JAVA7	−8.4	105	228.2	32.4	329	261	271
	SUM1	−7	102.3	348.9	33.5	220	168	159
	SUM2	−5.5	100.8	239.4	50.8	246	182	185
	SUM4	−2	98.1	265.2	47.3	446	318	450
	SUM5	0	97	314.5	56	398	298	394
	SUM6	2	95	366.8	25	318	254	278
TK	HIKN2	−36.5	180.6	230.2	292	228	206	232

Table 1. (continued)

Zone	Profile	Latitude	Longitude	Width (km)	Azimuth (deg)	R_{min} (km)	R_{vs} (km)	R_{pl} (km)
	HIKS1	−42.3	175	314.5	320	305	258	314
	HIKS2	−41.4	177	314.5	332	432	293	427
	HIKS3	−40	178.7	225.8	287	296	274	300
	KER1	−35	181.6	200.5	292.5	134	136	134
	KER2	−33	182.2	206.4	287.9	164	161	156
	KER3	−31	183.2	213.9	291.1	120	116	106
	KER4	−29	183.9	212	286.4	158	140	158
	KER5	−27	184.5	201.9	281.2	202	99	70
	TONG1	−25	184.8	202.4	275.9	249	187	173
	TONG2	−23	185.4	227.2	297.9	180	137	107
	TONG3	−21	186.5	215.8	291.1	185	98	69
	TONG4	−19	187.2	208.3	286.8	195	133	110
	TONG5	−17	187.8	204.4	286.1	131	103	73
	TONG6	−15.5	187.7	221.9	241.4	100	68	51

[a]Profile names are taken from *Heuret and Lallemand* [2005] with locations indicated by latitude and longitude at the trench.

p clearly yields a poor fit to the seismicity at shallow depths. The highest value for p gives R_{min} = 143 km, whereas the intermediate value ($p = 0.5 \times 10^{-6}$) gives R_{min} = 164 km. The difference (≈15%) is a plausible estimate for the uncertainty in the fit. A modified procedure is applied to vertically dipping slabs because the depth z is not uniquely specified by the horizontal position x. In this case we rotate the x-z coordinates of the hypocenters through a small angle (15 degrees) before applying the spline fit. This transformation improves the reliability of the fit for steeply dipping slabs, but it has little affect on the results for profiles like that in the central Aleutian (Figure 1). For the sake of consistency we apply the same transformation to all profiles.

3. Radius of Curvature and Related Quantities

[9] Bending dissipation in the slab is determined by changes in curvature with distance along the slab.

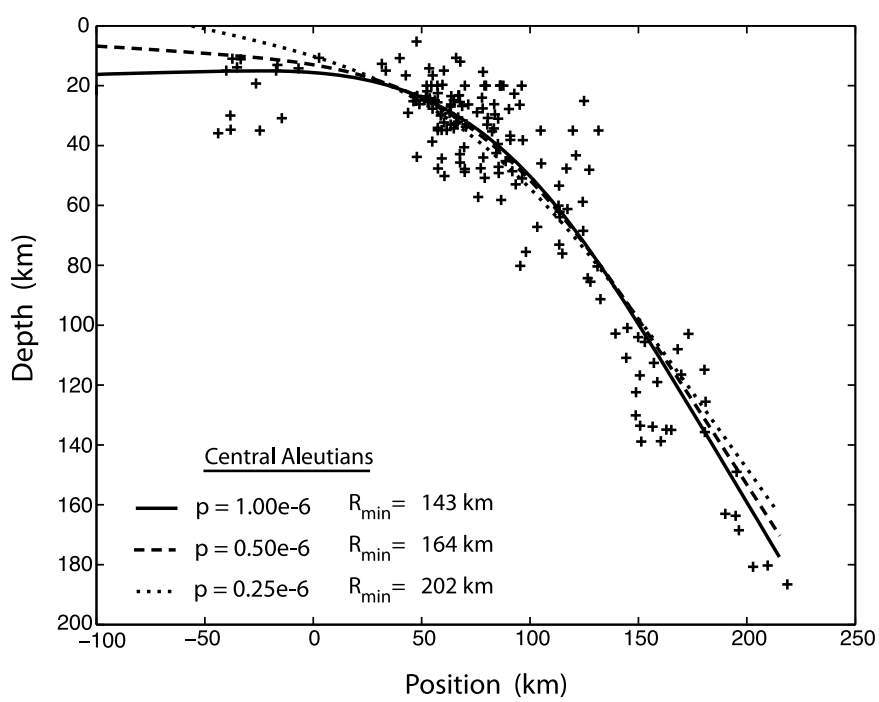

Figure 1. A representative spline fit through hypocenters in the central Aleutian subduction zone. Three different values are chosen for the smoothing parameter p. A better fit to the hypocenters is obtained with the largest value ($p = 10^{-6}$). The minimum radius of curvature, R_{min}, is calculated for each of the spline fits.

For a thin viscous sheet the bending dissipation is given by *Buffett* [2006] as

$$\Phi = \frac{1}{3}\eta u^2 H^3 \int_0^L \left(\frac{dK}{ds}\right)^2 ds \tag{2}$$

where η is the viscosity of the plate, u is the normal velocity of the downgoing plate relative to the trench, H is the thickness of the plate and L is the total length of the slab. The derivation of (2) assumes that the shape of the slab in a reference frame fixed to the trench does not change with time. This assumption applies to a wide range of conditions, from steady subduction at a fixed trench to the case of rapid trench rollback, provided the curvature of the slab relative to the moving trench location does not change with time. Newly formed subduction zones could evolve with time so that the curvature relative to the trench may not be constant. However, the deformation due to a broad time-dependent change in curvature (either increasing or decreasing curvature) would only increase the dissipation. In this case, the dissipation predicted by (2) represents a lower bound. While we cannot rule out the possibility that spatial variations in the time dependence could lower the dissipation, this behavior is not observed in laboratory [*Schellart*, 2009] or numerical [*Capitanio et al.*, 2009] models of subduction zones.

[10] An approximation to (2) is obtained by assuming that the maximum curvature, $K = R_{\min}^{-1}$, develops over a distance which is comparable to $R_{\min}$. In this approximation

$$\frac{dK}{ds} \approx \frac{1}{R_{\min}^2} \tag{3}$$

and the limit of integration is $L \approx 2R_{\min}$. The resulting expression for the bending dissipation simplifies to

$$\Phi \approx \frac{2}{3}\left(\frac{H^3}{R_{\min}^3}\right)\eta u^2 \tag{4}$$

It is preferable to use the spline fits to integrate (2) directly for the bending dissipation. The result of the integration can be represented in terms of an equivalent radius of curvature, defined by

$$R_{vs} = 2^{1/3}\left[\int_0^L \left(\frac{dK}{ds}\right)^2 ds\right]^{-1/3} \tag{5}$$

An estimate for Φ in (2) is obtained by substituting R_{vs} for $R_{\min}$ in (4).

[11] Another geometric quantity, R_{pl}, is defined for a plastic plate. The strength of oceanic lithosphere is limited by brittle failure in the upper part of the plate and ductile creep below. A typical profile of yield stress increases with depth in the brittle zone due to increasing normal stress [*Byerlee*, 1978]. In the ductile zone the yield stress decreases with depth because temperature and strain rate are higher. We consider a simple plastic model in which the yield stress, $\tau_y(z)$, is maximum on the centerline ($z = 0$) and decreases linearly to zero at the top and bottom of the plate. A nominal value for the yield stress on the centerline is $\tau_y(0) = 600$ MPa, corresponding to a depth-averaged yield stress of 300 MPa [*Mei et al.*, 2010]. When a single yield stress envelope is applied over the entire length of the slab, the bending dissipation is given by *Buffett* [2006] as

$$\Phi = \frac{1}{12}\tau_y(0) u H^2 \int_0^L \left|\frac{dK}{ds}\right| ds \tag{6}$$

where $|\ |$ denotes the absolute value. Approximating (6) using (3) gives

$$\Phi \approx \frac{1}{6}\left(\frac{H^2}{R_{\min}}\right)\tau_y(0) u \tag{7}$$

In analogy with the viscous sheet, we define an equivalent radius of curvature for a plastic sheet using

$$R_{pl} = 2\left[\int_0^L \left|\frac{dK}{ds}\right| ds\right]^{-1} \tag{8}$$

We obtain an estimate for Φ in (6) by substituting R_{pl} for $R_{\min}$ in (7). Note that the plastic model for Φ depends linearly on u, whereas the viscous model has a quadratic dependence.

[12] For a representative profile from the central Aleutians (Figure 1), we obtain $R_{\min} = 143$ km, $R_{vs} = 131$ km and $R_{pl} = 143$ km using a smoothing of $p = 10^{-6}$. (Somewhat larger values $R_{\min} = 164$ km, $R_{vs} = 147$ km and $R_{pl} = 165$ km are obtained with $p = 0.5 \times 10^{-6}$.) Rough agreement between $R_{\min}$ and the equivalent radius (R_{vs} and R_{pl}) means that the approximate expressions for Φ in (4) and (7) are reasonably good. This is not the case for more complicated profiles of seismicity with multiple bends (like those under South America). In these locations R_{vs} or R_{pl} should be used in place of $R_{\min}$ in (4) or (7) to estimate the bending dissipation.

[13] *Heuret* [2005] measured the radius of curvature of slabs using the same transects and hypocenters,

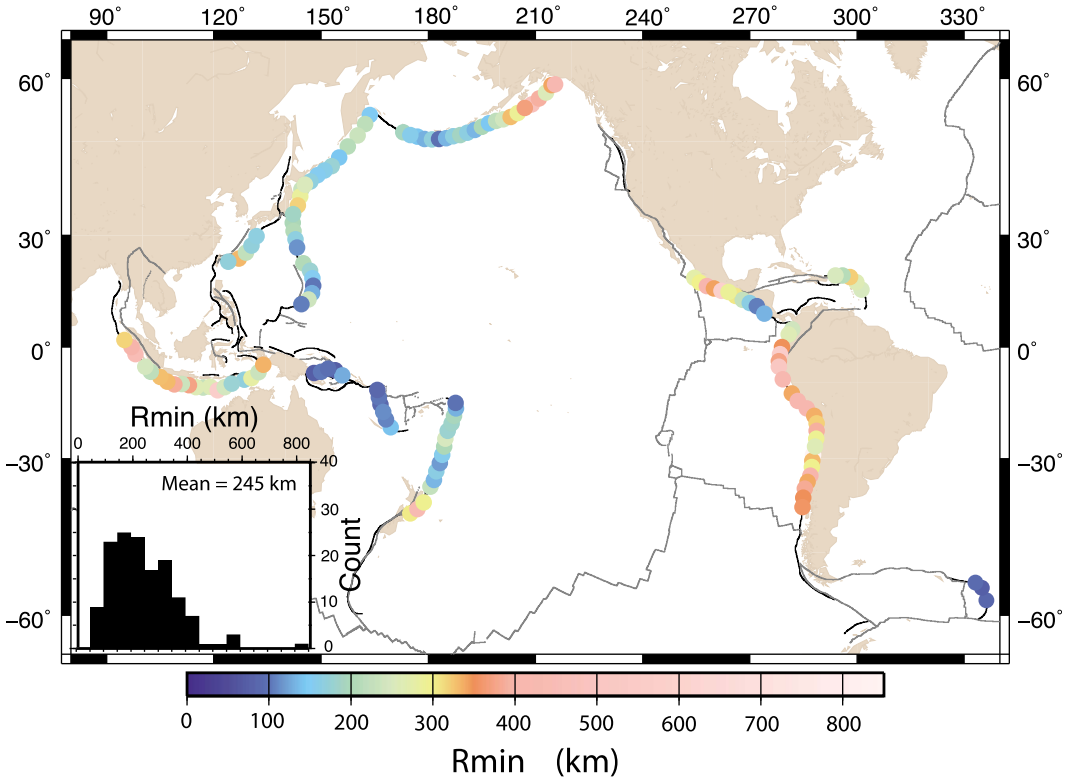

Figure 2. Map of the profile locations. The color code denotes the minimum radius of curvature, $R_{\min}$, computed from spline fits to the earthquake hypocenters. Inset is a histogram of values for $R_{\min}$ giving a global mean of $R_{\min}$ = 245 km.

but employed a different measurement strategy. A circle was fit to the outer envelope of seismicity from the surface to a depth of roughly 150 km, defining an average radius of curvature along the top surface of the slab. A slightly smaller radius of curvature is expected for the centerline, but the main difference with the results presented here is the averaging of curvature through the bend. Curvature in a slab typically increases to a maximum value, $K = R_{\min}^{-1}$, before decreasing again as the slab straightens. An average radius of curvature is expected to be roughly twice as large as $R_{\min}$. More importantly, the average is larger than the equivalent radius of curvature, which characterizes the bending dissipation in a viscous (5) or plastic (8) sheet. For example, the average radius of curvature for profile C_ALE1 is 240 km [*Wu et al.*, 2008], whereas the equivalent radius is 131 km to 143 km (for $p = 10^{-6}$).

[14] Figure 2 shows the location of profiles used in this study. Each location is color coded according to the value of $R_{\min}$. (Maps of R_{vs} and R_{pl} are shown in Figures 3 and 4, respectively.) We retain only a subset of the profiles defined in *Heuret and Lallemand* [2005] by limiting our attention to major subduction zones and by excluding profiles with insufficient seismicity to reliably fit splines. In many instances the profiles with insufficient seismicity have larger radii of curvature, so our sample may be biased. However, this bias is probably small because our mean value is roughly half (i.e., within expectations) of the mean value from the more complete sample of *Wu et al.* [2008]. A detailed list of the profiles and corresponding values for $R_{\min}$, R_{vs} and R_{pl} are given in Table 1. The global mean and standard deviation are $R_{\min}$ = 245 ± 120 km, R_{vs} = 175 ± 60 km, and R_{pl} = 191 ± 90 km. By comparison, the mean value reported by *Wu et al.* [2008] is R = 390 km.

[15] Regional averages of R_{vs} and R_{pl} are shown as a function of plate age in Figure 5. Each point represents an average of several profiles from a particular subduction zone. The error bar indicates the standard deviation among the individual profiles in a given average. Most of the variability is attributed to variations along the subduction boundary. The corresponding standard deviation for the average age is not shown for clarity. The results show very little dependence on the age of the plate. In fact,

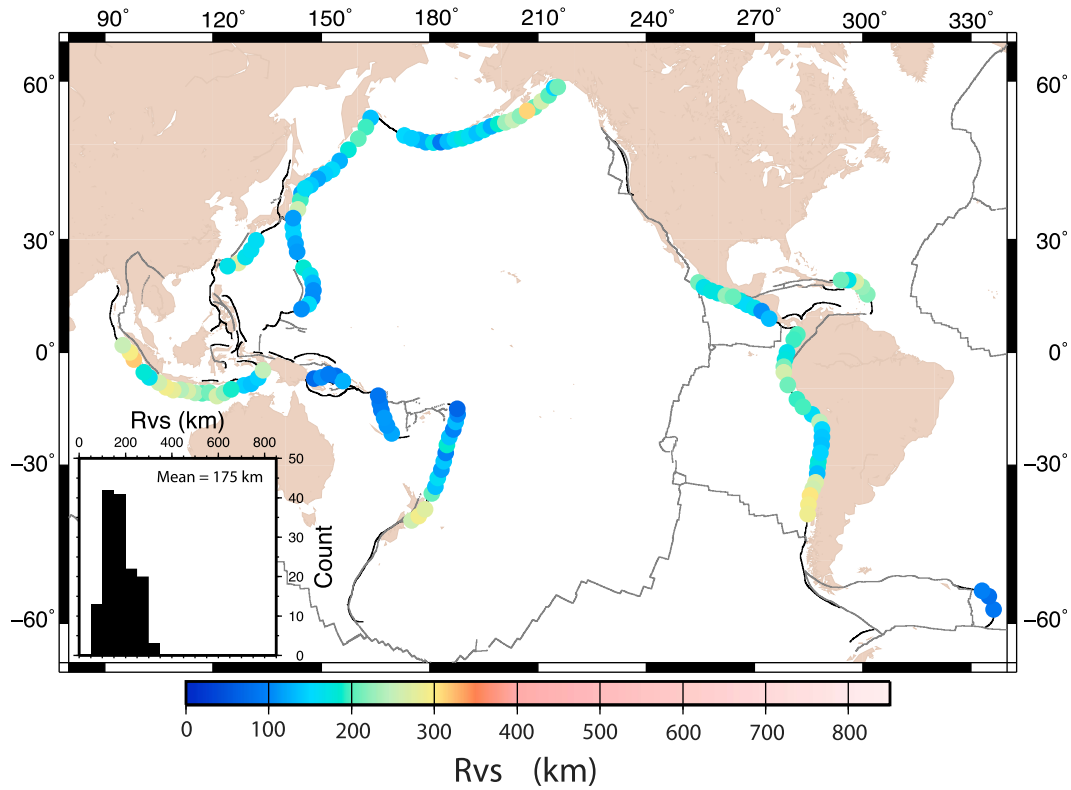

Figure 3. Map of the profile locations. The color code denotes the equivalent radius of curvature, R_{vs}, computed from spline fits to the earthquake hypocenters. Inset is a histogram of values for R_{vs} giving a global mean of R_{vs} = 175 km.

the oldest plates (Tonga/Kermadec, Japan/Kuriles and Marianas) have a somewhat smaller radius of curvature than the mean values. We now examine the consequences for bending dissipation.

4. Dissipation due to Bending

[16] Slab shape is only one of the factors that determine the dissipation due to bending. Others include the speed and rheological properties of the subducting plate. An equivalent radius of curvature (either R_{vs} or R_{pl}) provides a general description of the slab shape, while the velocity of subducting plates relative to the trench (sometimes called the subduction velocity) is taken from plate motion models [*Gripp and Gordon*, 2002]. More specifically, the bending dissipation depends on the normal-to-trench component of the subduction velocity [*Lallemand et al.*, 2005]. Here we use the subduction velocities and ages tabulated by *Wu et al.* [2008]. The rheological properties are less certain. Nominal values for slab viscosity ($\eta = \times 10^{22}$ Pa s) and yield stress ($\tau_y(0)$ = 600 MPa) suffice to illustrate the consequences of slab shape, as inferred from the earthquake locations.

[17] Bending of oceanic plates at subduction zones opposes the plate motions. Part of the energy required to overcome this resistance is supplied by the work of cold slabs sinking through the upper mantle. A buoyancy flux is calculated using

$$B = \overline{\Delta\rho} g H D u \tag{9}$$

where $\overline{\Delta\rho}$ = 65 kg m^{-3} is the average density excess of the slab relative to the mantle, g = 9.8 m s^{-2} is the acceleration due to gravity and D = 670 km is the depth of the upper mantle. The thickness H of the oceanic lithosphere is calculated as a function of age using the depth of the ocean and the assumption of isostasy [*Parsons and Sclater*, 1977]. The ocean depth is fit to a half-space cooling model [*Carlson and Johnson*, 1994] for ages less than 81 Ma and to a simple parametric function for ages greater than 81 Ma [*Parsons and Sclater*, 1997]. The same value for H is used in calculating the bending dissipation, although this assumption is open to question. If the correct value for H in (2) or (6) is proportional to the thermal thickness, then the constant of proportionality could be absorbed in the definition of the rheological constant (either η or $\tau_y(0)$) without changing the relative amplitude of the results pre-

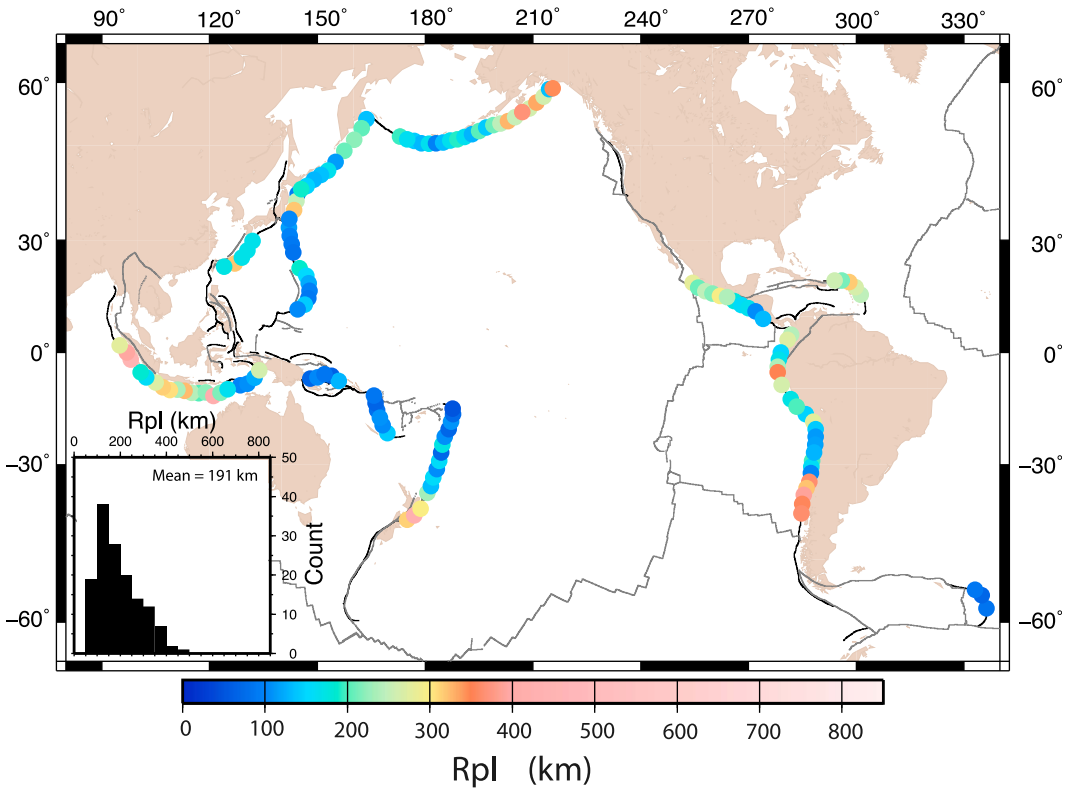

Figure 4. Map of the profile locations. The color code denotes the equivalent radius of curvature, R_{pl}, computed from spline fits to the earthquake hypocenters. Inset is a histogram of values for R_{pl} giving a global mean of R_{pl} = 191 km.

sented below. However, any interpretation of the rheological constants is dependent on the definition of H.

[18] Figure 6 shows the ratio Φ/B for both viscous and plastic slabs. Each point represents an average of profiles from a particular subduction boundary. Individual estimates of Φ and B are used to evaluate Φ/B for every profile. An average of Φ/B and the plate age yields a single result for a specific boundary region; the error bars indicate the standard deviation about the mean. Surprisingly large values for Φ/B are obtained in viscous slabs. Indeed, the average ratio for Tonga-Kermadec and New Hebrides exceeds unity, which implies that the energy dissipated by bending is greater than the energy available from sinking slabs. Several of the individual profiles in Tonga-Kermadec and New Hebrides have ratios that substantially exceed unity, causing large standard deviations. Other regions (Marianas, New Britain and Sandwich) have average ratios that exceed 0.8, with a few individual profiles exceeding 1. Interestingly, all of these subduction zones were initiated within the last 45 Ma [*Gurnis et al.*, 2004]. If these slabs are still evolving toward higher (or lower) curvatures, we might expect even higher levels of dissipation due to the additional deformation.

[19] A ratio in excess of unity is not strictly prohibited because plates can be driven by basal stress due to buoyancy elsewhere in the mantle. However, it is unlikely for a broad region of the subduction boundary to have a bending dissipation that exceeds the buoyancy flux. This claim is supported by inferences from the direction of plate motions, suggesting that $\Phi/B \leq 0.36$ [*Buffett and Rowley*, 2006]. Similarly, *Wu et al.* [2008] argue for $\Phi/B < 0.25$, based on the speed of plate motions. While numerical models of subduction can predict $\Phi/B > 0.3$ when the slab is strong [*Conrad and Hager*, 2001; *Di Giuseppe et al.*, 2008; *Capitanio et al.*, 2007], most simulations of mantle convection report much smaller values for Φ/B [e.g., *Stadler et al.*, 2010]. Some caution is required in making comparisons because the definition of B in mantle convection studies usually refer to the buoyancy flux across the entire mantle [*Buffett*, 2010].

[20] Large values for Φ/B in a viscous slab might imply that the nominal slab viscosity is too high or that the plate thickness (for bending) is too large. Lowering the slab viscosity by a factor of 4 (or reducing the thickness by a factor of $4^{1/3}$) brings the dissipation for Tonga-Kermadec into line with previous inferences. However, a large reduction

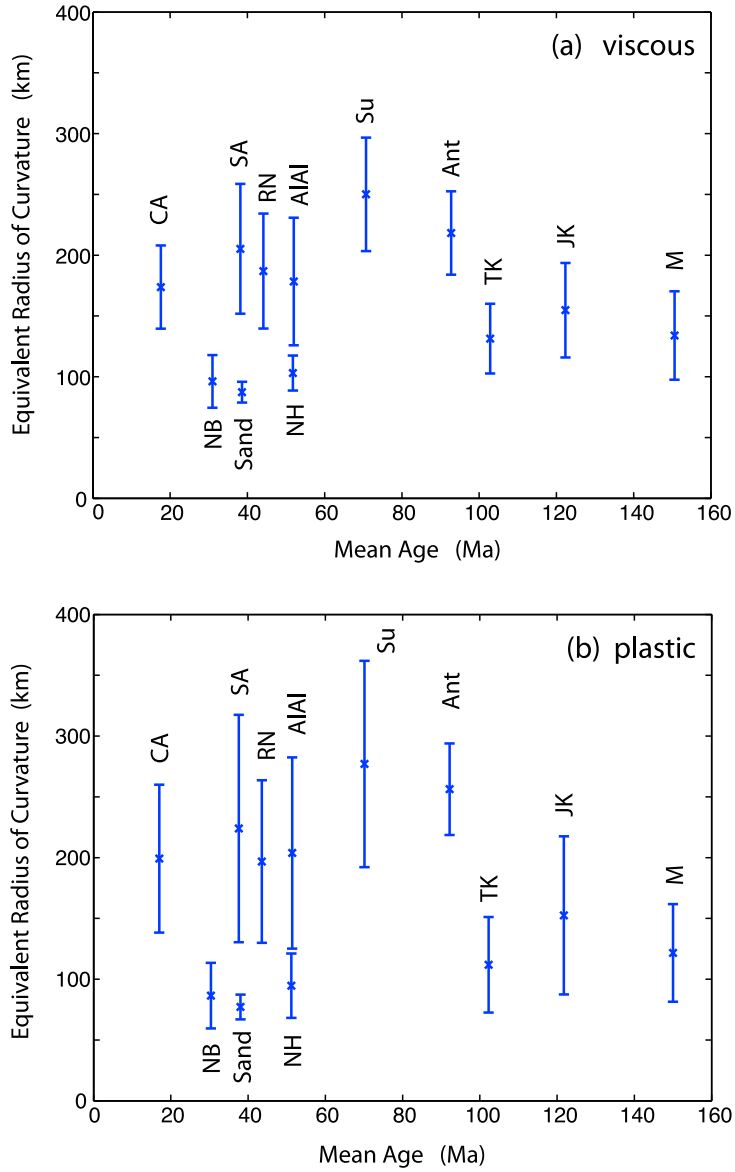

Figure 5. Equivalent radius of curvature for (a) viscous and (b) plastic slabs. Each point represent an average of several profiles at a particular subduction zone. The error bar gives the standard deviation among the individual profiles in an average. The subduction zones are CA, Central America; SA, South America; NB, New Britain; Sand, Sandwich; RN, Ryukyu/Nankai; AlAl, Alaska/Aleutian; NH, New Hebrides; Su, Sunda; Ant, Antilles; TK, Tonga/Kermadec; JK, Japan/Kuriles; and M, Marianas.

in viscosity can also reduce the integrity of the slab, limiting the transmission of slab pull to the surface [*Capitanio et al.*, 2007]. *Wu et al.* [2008] favor a slab viscosity of 2×10^{23} Pa s to ensure that slab pull has a strong influence on plate motions [*Conrad and Lithgow-Bertelloni*, 2002]. Such a viscosity is nearly 7× larger than the nominal value in this study, implying that Φ/B exceeds unity at most subduction zones.

[21] Alternatively, the large bending dissipation in viscous slabs might suggest that slabs do not deform viscously. The largest values for Φ/B occur when the radius of curvature is small and the subduction velocity is large. A plastic rheology does not substantially change the equivalent radius of curvature. However, the bending dissipation in a plastic slab has a much weaker dependence on the subduction velocity. As a consequence, Φ/B in a plastic slab never exceeds 40% for a plausible yield stress.

[22] *Wu et al.* [2008] sought to reconcile a weak bending force with strong slab pull by increasing the radius of curvature above the commonly adopted

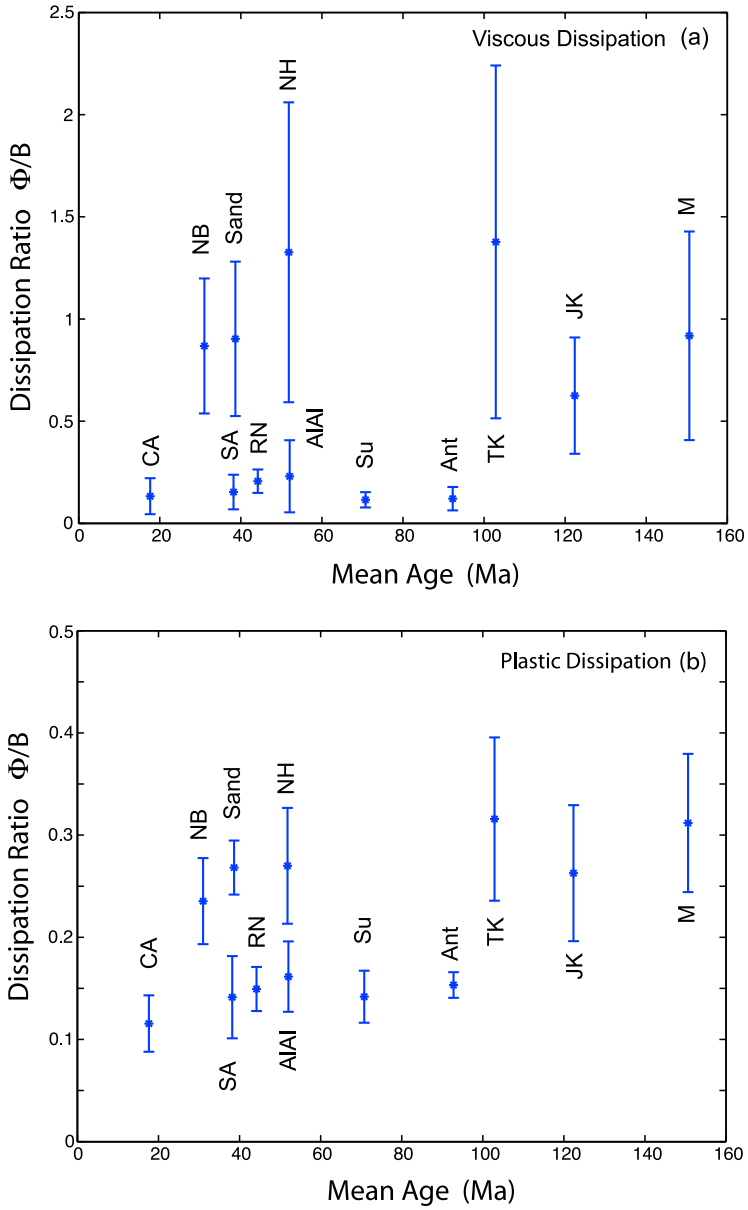

Figure 6. Ratio of bending dissipation Φ and buoyancy flux B as a function of plate age at the trench for (a) viscous and (b) plastic slabs. Note the differences in the range of values on the vertical axis. The viscous dissipation is calculated using an average slab viscosity of $\eta = 3 \times 10^{22}$ Pa s and the plastic dissipation is calculated with a peak yield stress of $\tau_y(0) = 600$ MPa, corresponding to a depth-averaged yield stress of 300 MPa. The bending dissipation in a viscous slab can exceed the buoyancy flux, whereas the dissipation in a plastic slab accounts for no more than 40% of the buoyancy flux.

value of $R = 200$ km [e.g., *Bevis*, 1986]. Our spline fit to hypocenters indicates that $R \approx 200$ km is a reasonable estimate for the mean. In fact, the radius of curvature for the oldest slabs is even smaller. This weak trend in curvature as a function of thickness is not expected from laboratory [*Bellahsen et al.*, 2005] and numerical [*Capitanio et al.*, 2009] studies. While a larger bending moment is expected in older slabs [*Lallemand et al.*, 2008], the resulting stress must produce substantially larger strain rates to keep the radius of curvature nearly constant with increasing age or thickness. On energetic grounds a balance between B and Φ in a viscous slab requires R_{vs} to vary with thickness as $H^{2/3}$ when the plate speed u is weakly dependent on age. The fact that R_{vs} is relatively independent of H suggests that either a viscous rheology is inappropriate or that the bending dissipation is sufficiently small to relax

the constraint imposed by energy conservation (i.e., $\Phi \approx B$). Requiring $\Phi \ll B$ in a viscous slab means that $\eta \ll 3 \times 10^{22}$ Pa s because the nominal viscosity yields $\Phi \approx B$ for a range of ages (see Figure 3). A simple plastic model gives a more consistent description because the bending dissipation is unlikely to exceed 40% of B, even for old oceanic lithosphere. As long as Φ is less than B we do not require R_{pl} to vary with H on the basis of energy conservation. Therefore a plastic rheology appears to be compatible with the nearly constant radius of curvature. On the other hand, we can not rule out the possibility that systematic variations in R with age are obscured by natural variability in other slab properties, including the length, density, width, and viscosity of the slab [e.g., *Faccenna et al.*, 2007].

Acknowledgments

[23] We thank Mike Gurnis, Clint Conrad and an anonymous reviewer for many constructive comments that improved the paper. This work is partially supported by the National Science Foundation (EAR-0911255). A.H. participation was supported as part of the Eurohorcs/ESF—European Young Investigators Awards Scheme (Francesca Funiciello), by funds from the National Research Council of Italy and other National Funding Agencies participating in the 3rd Memorandum of Understanding, as well as from the EC Sixth Framework Programme.

References

Barazangi, M., and B.L. Isacks (1976), Spatial distribution of earthquakes and subduction of the Nazca plate beneath South America, *Geology*, *4*, 686–692.

Becker, T. W., C. Faccenna, R. J. O'Connell, and D. Giardini (1999), The development of slabs in the upper mantle: Insights from numerical and laboratory experiments, *J. Geophys. Res.*, *104*, 15,207–15,226, doi:10.1029/1999JB900140.

Bellahsen, N., C. Faccenna, and F. Funiciello (2005), Dynamics of subduction and plate motion in laboratory experiments: Insights into the "plate tectonics" behavior of the Earth, *J. Geophys. Res.*, *110*, B01401, doi:10.1029/2004JB002999.

Bevis, M. (1986) The curvature of Wadati-Benioff zones and the torsional rigidity of subducting plates, *Nature*, *323*, 52–53.

Bevis, M., and B. L. Isacks (1984), Hypocentral Trend Surface Analysis: Probing the Geometry of Benioff Zones, *J. Geophys. Res.*, *89*, 6153–6170, doi:10.1029/JB089iB07p06153.

Boyd, T. M., and K. C. Creager (1991), The geometry of Aleutian subduction: Three-dimensional seismic imaging, *J. Geophys. Res.*, *96*, 2267–2291, doi:10.1029/90JB01919.

Buffett, B. A. (2006), Plate force due to bending at subduction zones, *J. Geophys. Res.*, *111*, B09405, doi:10.1029/2006JB004295.

Buffett, B. A. (2010), Comment on "Evolution of the slab bending radius and bending dissipation in three dimensions with variable slab to upper mantle viscosity ratio" by W. P. Schellart, *Earth and Planetary Science Letters, 288* (2009) 309–319, *Earth Planet. Sci. Lett.*, *296*, 153–154.

Buffett, B. A., and D. B. Rowley (2006), Plate bending at subduction zones: Consequences for the direction of plate motions, *Earth Planet. Sci. Lett.*, *245*, 359–364.

Byerlee, J. D. (1978), Friction of rocks, *Pure Appl. Geophys.*, *116*, 615–626.

Cahill, T., and B. L. Isacks (1992), Seismcity and shape of the Nazca plate, *J. Geophys. Res.*, *97*, 17,503–17,529.

Capitanio, F. A., G. Morra, and S. Goes (2007), Dynamic models of downgoing plate-buoyancy driven subduction: Subduction motions and energy dissipation, *Earth Planet. Sci. Lett.*, *262*, 284–297.

Capitanio, F. A., G. Morra, and S. Goes (2009), Dynamics of plate bending at the trench and slab-plate coupling, *Geochem. Geophys. Geosyst.*, *10*, Q04002, doi:10.1029/2008GC002348.

Carlson, R. L., and H. P. Johnson (1994), On modeling the thermal evolution of the oceanic upper mantle: An assessment of the cooling plate model, *J. Geophys. Res.*, *99*, 3201–3214.

Chiu, J. M., B. L. Isacks, and R. K. Cardwell (1991), 3-D configuration of subducted lithosphere in the western Pacific, *Geophys. J. Int.*, *106*, 99–111.

Conrad, C. P., and B. H. Hager (1999), The effects of plate bending and fault strength at subduction zone on plate dynamics, *J. Geophys. Res.*, *104*, 17,551–17,571.

Conrad, C. P., and B. H. Hager (2001), Mantle convection with strong subduction zones, *Geophys. J. Int.*, *144*, 271–288.

Conrad, C. P., and C. Lithgow-Bertelloni (2002), How mantle slabs drive plate tectonics, *Science*, *298*, 207–209.

Di Giuseppe, E., J. van Hunen, F. Funiciello, C. Faccenna, and D. Giardini (2008), Slab stiffness control of trench motion: Insights from numerical models, *Geochem. Geophys. Geosyst.*, *9*, Q02014, doi:10.1029/2007GC001776.

Engdahl, R., R. van der Hilst, and R. Burland (1998), Global teleseismic earthquake relocation with improved travel times and procedures for depth determination, *Bull. Seismol. Soc. Am.*, *88*, 722–743.

Faccenna, C., A. Heuret, F. Funiciello, S. Lallemand, and T. W. Becker (2007), Predicting trench and plate motion from the dynamics of a strong slab, *Earth Planet. Sci. Lett.*, *257*, 29–36.

Forsyth, D.W. (1982), Determination of focal depths of earthquakes associated with the bending of oceanic plates at trenches, *Phys. Earth Planet. Inter.*, *28*, 141–160.

Gripp, A. E., and R. G. Gordon (2002), Young tracks of hotspots and current plate velocities, *Geophys. J. Int*, *150*, 321–361.

Gurnis, M., C. Hall, and L. Lavier (2004), Evolving force balance during incipient subduction, *Geochem. Geophys. Geosyst.*, *5*, Q07001, doi:10.1029/2003GC000681.

Heuret, A. (2005), Dynamique des zone de subduction: Etude statistique globale et approache anologique, Ph.D. thesis, Univ. Montellier II, Montpellier, France.

Heuret, A., and S. Lallemand (2005), Plate motions, slab dynamics and back-arc deformation, *Phys. Earth Planet. Inter.*, *149*, 31–51.

Lallemand, S., A. Heuret, and D. Boutelier (2005), On the relationship between slab dip, back-arc stress, upper plate motion and crustal nature in subduction zones, *Geochem. Geophys. Geosyst.*, *6*, Q09006 doi:10.1029/2005GC000917.

Lallemand, S., A. Heuret, C. Faccenna, and F. Funiciello (2008), Subduction dynamics as revealed by trench migration, *Tectonics*, *27*, TC3014, doi:10.1029/2007TC002212.

Mei, S., A. M. Suzuki, D. L. Kohlstedt, N. A. Dixon, and W. B. Durham (2010), Experimental constraints on the

strength of lithospheric mantle, *J. Geophys. Res.*, *115*, B08204, doi:10.1029/2009JB006873.

Molnar, P., D. Freedman, and J. S. F. Shih (1979), Lengths of intermediate and deep seismic zones and the temperature in downgoing slabs of lithosphere, *Geophys. J. R. Astron. Soc*, *56*, 41–54.

Nothard, S., D. McKenzie, J. Haines, and J. Jackson (1996), Gaussian curvature and the relationship between the shape and the deformation of the Tonga slab, *Geophys. J. Int*, *127*, 311–327.

Pardo, M., and G. Suarez (1995), Shape of the subducted Rivera and Cocos plates in southern Mexico: Seismic and tectoni implications, *J. Geophys. Res.*, *100*, 12,357–12,373.

Parsons, B., and J. G. Sclater (1977), Analysis of variation of ocean-floor bathymetry and heat flow with age, *J. Geophys. Res.*, *82*, 803–827.

Schellart, W. P. (2009) Evolution of the slab bending radius and bending dissipation in three dimensions with variable slab to upper mantle viscosity ratio, *Earth Planet. Sci. Lett.*, *288*, 309–319.

Stadler, G., M. Gurnis, C. Burstedde, L. C. Wilcox, L. Alisic, and O. Ghattas (2010), The dynamics of plate tectonics and mantle flow: From local to global scales, *Science*, *329*, 1033–1038.

Stauder, W. (1973), Mechanism and spatial distribution of Chilean earthquakes with relation to subduction of oceanic lithosphere, *J. Geophys. Res.*, *78*, 5033–5061.

Wu, B., C. P. Conrad, A. Heuret, C. Lithgow-Bertelloni, and S. Lallemand (2008), Reconciling strong slab pull and weak plate bending: The plate motion constraint on mantle slabs, *Earth Planet. Sci. Lett.*, *272*, 412–421.